

Tonal variation and word meaning in Taiwanese

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Abstract

Tone sandhi is extensive in Taiwanese, where all but the final syllable in a morphosyntactically defined unit undergo sandhi. Previous studies have primarily focused on comparing sandhi tonal realizations to their corresponding citation ones to determine whether neutralization occurs. However, such a high-level comparison overlooks the variability in how tones are actually realized. This study aims to examine tonal variation in Taiwanese, focusing on whether word meaning influences tonal realization. Using spontaneous speech data, we analyzed the realizations of the high-falling tone. Our findings indicate that word meaning does contribute to tonal variability. Moreover, when meaning-induced tonal variation is accounted for, no remaining differences between sandhi and citation tones can be observed. These results suggest that tonal realizations systematically vary with word meaning and that the effect of meaning on tonal realizations should be considered when discussing neutralization.

Index Terms: tonal variation, word meaning, tone sandhi, tonal neutralization, Taiwanese

1. Introduction

Taiwanese, also commonly known as Taiwan Southern Min, is the second widely spoken language in Taiwan, after Mandarin. Similar to Mandarin, Taiwanese is also a tone language, but its tonal inventory is richer and the tone sandhi system is more complex. The dominant dialect of Taiwanese is often described to have seven lexical tones, including five long (smooth) tones and two short (checked) tones. The distinction between long and short tone is based on syllable structure, with the short tones restricted to syllables with stop codas. The sandhi system in Taiwanese, regardless of dialects, is continuous. That is, sandhi applies to every non-final syllable. Thus, a syllable's citation tone only occurs at boundaries of units that are mostly morphosyntactically defined, and all the other syllables should be realized with sandhi tones. For example, the words *to21* 'pour' and *tsui51* 'water' have a low-falling and a high-following underlying tone respectively. When they are put together to form the phrase of 'pouring water', the first word undergoes sandhi and is realized as a high-falling tone, while the second word, if occurring before a boundary, is still realized as a high-falling tone. That is, despite the same high-falling tone, the former realizes its sandhi tone, whereas the latter its citation tone.

The realization of sandhi tone has attracted a lot of attention. Traditionally, it has been considered that the sandhi realization (e.g., low-falling realized as high-falling) is indistinguishable from the realization of the corresponding citation tone (e.g., high-falling realized as high-falling). In other words, the sandhi process involves categorical tonal alternation [1, 2], lead-

ing to tonal neutralization. More recent studies, however, suggest the contrary. Particularly with spontaneous speech data, it was found that sandhi and citation tonal realizations are not identical [3, 4]. This indicates that the two tonal realizations are incompletely neutralized. Furthermore, it was even shown that perceptually listeners are able to tell the difference between sandhi and citation realizations at above-chance level [5, 3].

As considerable discussion has been devoted to the issue of tonal neutralization, so far little attention has been given to the actual variability present in tonal realizations. That is, comparisons between sandhi and citation realizations to a large extent remain at a broad level, and to the best of our knowledge, no study has directly addressed the variation involved in sandhi and citation tonal realizations. In fact, it is well established, at least for Mandarin, that tonal realizations vary substantially in connected speech, and may even lose their tonal shapes denoted by the canonical tonal descriptions [6, 7]. Furthermore, various factors, such as coarticulation, syllable structure, intonation, prosodic prominence, and emotion, can all affect how lexical tones are realized, contributing to tonal variation [8].

The goal of this study is to investigate tonal variation in sandhi and citation tonal realizations in Taiwanese. Specifically, we are interested in whether tonal variation is to some extent lexically determined. That is, does word meaning, along with other known factors, co-determine tonal realizations in Taiwanese? This research question arises from a growing body of research showing that fine-grained phonetic realizations vary with word meaning. For example, [9] reported that English heterographic homophones such as *time* and *thyme* differ consistently in duration. Moreover, different senses of the same word (e.g., *like* used as a verb or a discourse particle) also have distinct phonetic realizations [10]. At the sublexical level, [11] found that the English /s/ suffix is produced with varying durations depending on its morphological functions (e.g., plural, third-person singular). Given the intimate relationship between what a word means and how it is realized phonetically, following this line of research, in this study we hypothesize that different words, homophonous or not, should also differ in their tonal realizations.

To address this issue, we make use of the Taiwan Southern Min spontaneous speech corpus [12]. In this study, we focus on monosyllabic words whose surface realization is a high-falling tone, which has the most evenly distributed tokens across various conditions in our data. The high-falling realization can be either sandhi or citation tone: the former is the sandhi realization of words whose underlying tone is low-falling, whereas the latter is the citation realization of words whose underlying tone is already high-falling. As sandhi and citation realizations are usually found to be incompletely neutralized in spontaneous speech, for the current analyses we also expect to find

non-identical realizations for sandhi and citation realizations. However, given that no previous studies have considered the effect of word meaning, it remains unknown whether incomplete neutralization can still be observed after meaning-induced tonal variation is taken into account.

As variation usually manifests in fine-grained phonetic details, in this study, instead of only using measures gauged tonal realizations at a grand level such as pitch height or range, our analyses are conducted on the entire pitch contours. Specifically, we analyze tonal contours by using the Generalized Additive Model [13, GAM]. As pitch varies with time, by predicting pitch from time, the smooth functions in GAM then are able to capture the undulations of pitch for the entire word. Further application of GAM to tone or pitch analyses, see for example [14] and [15].

2. Methods

2.1. Data

The Taiwan Southern Min corpus [12] includes recordings of 16 Taiwanese native speakers, each contributing approximately 30 minutes of interview speech. The speakers (half self-identified as female, and half as male) were between 20 and 60 years old at the time of recording. In total there are of 3,792 word types of 88,619 word tokens. About 42.6% of the words are monosyllabic words, among which 5,540 tokens are realized as high-falling tone. To ensure that each word type has sufficient tokens, we first removed word types with fewer than 20 tokens. This left us with a dataset of 4,294 tokens of 41 words. Twenty of the 41 words (1,423 tokens) are citation tone realizations, and 21 words (2,871 tokens) are sandhi tone realizations. That is, about half of the words have a high-falling tone as their underlying tone, while the other half undergo sandhi, changing from a low-falling tone to a high-falling tone. For each token, we then extracted the entire tonal contour by measuring F0 values at 21 equidistant normalized time points. This was done by making use of the ProsodyPro software [16], with the range of F0 estimation set to between 50 and 400 Hz.

2.2. Analyses

As a first step, we fitted a baseline model. Since the F0 distribution is positively skewed, we log-transformed the F0 values to approximate normality. To model the entire contour, we fitted a GAM to log-transformed F0s, with normalized time as the main predictor. We further included a by-smooth of sex to request a separate predicted contour for each sex. However, including this by-smooth did not result in model fit improvement, suggesting that male and female speakers realize high-falling tone similarly in terms of contour shape. We therefore removed the by-smooth for sex from our model, and only considered the intercept difference between males and females. A by-speaker factor smooth was included to take into account any potential speaker idiosyncrasy in realizing high-falling tone. We also entered a number of controlling variables into the baseline model, including token duration, adjacent tone, and syllable structure, as all of them have been previously identified to affect tonal realizations. In addition, given the effect of prosody on tonal realizations [17], for each token we also labeled whether it occurs at a prosodic boundary, i.e., whether it is the last word of an intonational phrase, following the criteria of the ToBI annotation system of Taiwanese [18]. It should be noticed that although utterance boundaries tend to coincide with prosodic boundaries, this is not necessarily so in spontaneous speech.

Table 1: *The number of word tokens and word types (in parenthesis) for citation and sandhi realizations that occur at pre-boundary and prosodic boundary positions.*

	pre-boundary	boundary
citation	1,002 (20)	421 (20)
sandhi	2,436 (21)	435 (21)

Table 1 presents the distribution of sandhi and citation tokens, cross-tabulated by their occurrence at the prosodic boundary. As expected, there are fewer citation than sandhi realizations at pre-boundary positions. There are, however, slightly more sandhi than citation realizations that occur at prosodic boundaries. These sandhi tokens primarily come from disfluencies and hesitation.

To examine whether citation and sandhi tones are neutralized, we then added the variable `type` (citation vs. sandhi) to the baseline model described above. If model fit, estimated by AIC [19], significantly improves with the addition of `type`, it then suggests that tonal realizations differ for sandhi and citation tones, thus indicating incomplete neutralization. On the contrary, if it does not lead to model fit improvement, this suggests that being informed about the source of tone type does not contribute to accounting for additional variance in the data, and it is therefore more likely that sandhi and citation tones are completely neutralized.

Finally, we investigated whether word meaning contributes to tonal variation, and if so, whether the effect of the word meaning is capable of explaining away the effect of `type`. For this set of analysis, we worked with a smaller dataset that contains only tokens that occur at pre-boundary positions. This is because the number of pre-boundary tokens outnumbered that of boundary ones substantially (cf. Table 1), unsurprisingly. Moreover, not all word types occur at prosodic boundaries. About a quarter of the word types have fewer than five boundary tokens, and one word never occurs at boundaries. Considering the effect of prosodic boundary, we therefore restricted the analysis of word meaning to pre-boundary tokens. We again first built a baseline model similar to the one described above with the smaller dataset, except that this time the variable `boundary` was not included. We then added a by-word factor smooth to the baseline model, requesting a predicted contour for each word type. Similar to the previous analyses, if this leads to model fit improvement, this suggests that word meaning, along with other known factors, also affect tonal realizations.

3. Results

In what follows, we first present results of citation and sandhi realizations, focusing on whether the two are completely or incompletely neutralized. Subsequently we show how tonal realizations differ by word, and finally, we discuss the realizations of three homophone pairs in the current data.

3.1. Citation and sandhi realizations

All the control variables in the baseline model significantly influence how tones are realized, consistent with the findings of previous studies on tonal variation. When the model is further informed about tone type (i.e., the `type` variable, citation vs. sandhi), model fit improves by 238 AIC units, suggesting that compared to the baseline model, the `type` model is $4.7978e + 51$ times more likely to be the generating model for

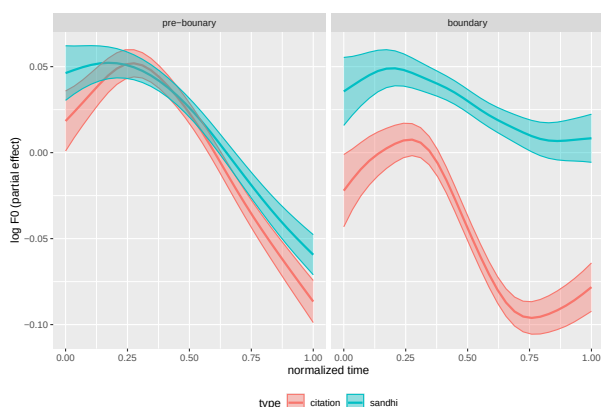


Figure 1: The predicted tonal contour (partial effect) for the citation and sandhi realizations that precede (left) and occur at (right) prosodic boundaries.

the current data. The predicted falling tone contours for citation and sandhi realizations at pre-boundary and boundary positions are shown in Figure 1. As can be seen, prosodic boundary has a strong effect on tonal realizations. The pre-boundary realizations start with a short period of level or rising, which is then followed by a fall. The boundary ones, on the other hand, generally have a shallower fall, and are accompanied by a small rise in the end. Importantly, tone type interacts with prosodic boundary. In pre-boundary positions, citation and sandhi realizations are more similar, but not identical. At prosodic boundaries, the difference is more pronounced, with sandhi realizations both higher and flatter than citation realizations. Moreover, the final rise is more prominent in citation realizations. This pattern of finding supports incomplete neutralization. On top of that, it further demonstrates how citation and sandhi realizations differ is subject to prosodic boundary.

3.2. By-word tonal variation

To investigate the effect of word meaning, we used a smaller dataset with only pre-boundary tokens, for reasons discussed previously in Section 2.2. We first fitted a baseline model with the same model structure as the previous analyses with the full dataset. All control variables fitted to this smaller dataset again also show significant effects. Crucially, by adding the by-word factor smooth to the baseline model, we observed substantial improvement of model fit, with a decrease of 980 AIC units. This suggests that *word* is a powerful predictor, which is capable of accounting for a large portion of variance in the data. Figure 2 visualizes the predicted tonal contours of a random selection of 12 words. Half of them are citation realizations (red), and the other half are sandhi realizations (green). While the falling contour seems to be present in all words, the detailed realizations however differ. The fall, for example, is rather muted in the words *be* ‘buy’ (panel [1, 1]) and *mai* ‘do not’ (panel [2, 4]). While the rising portion at the beginning is slow and extends for a longer period of time for *sui* ‘pretty’ (panel [1, 2]), it is however quick and short in *kio* ‘call’ (panel [2, 3]) and *tiam* ‘o’clock’ (panel [3, 1]). It can thus be seen that every word seems to have its own unique tonal realizations. Notably, although tonal realizations differ in a by-word fashion, there does not seem to be systematic differences between citation and sandhi tones.

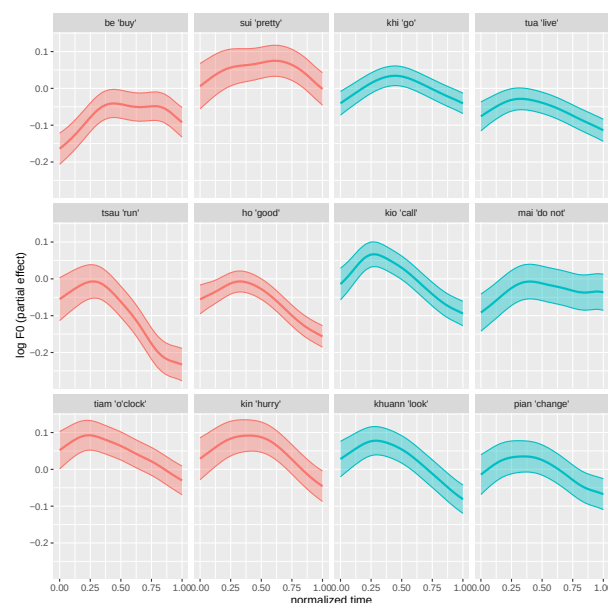


Figure 2: The predicted tonal contours for a sample of 12 words. The red and green lines indicate that the word is realized with citation and sandhi tone respectively.

To further delve into this issue, we fitted another model by adding the variable *type* to the baseline model. Compared to the baseline model, the *type* model was slightly better, as model fit improved by 5.45 AIC units. The extent of improvement, however, is by far smaller than that of the *word* model described above. Moreover, to investigate whether we can find any citation-sandhi difference beyond word-specific variation, we added *type* to the *word* model. Interestingly, model fit did not improve further. This suggests that once by-word tonal variation is taken into account, the information of tone type is no longer required. In other words, the small improvement that we observed for the *type* model might have resulted from some random similarities shared by words within the citation and the sandhi group. However, once by-word tonal variation is controlled for, no further differences between citation and sandhi tones can be observed.

3.3. Homophones

The analyses so far have confirmed the contribution of word in accounting for tonal variation. It however remains unknown whether the effect of word truly comes from word meaning. That is, given that different words also differ in various other aspects, it is possible that the effect of word results from not word meaning per se, but other factors such as segmental makeup and frequency of occurrence that also differ by word. To zoom in on the effect of word meaning, we thus went on to consider the tonal realizations of homophones.

In the current dataset, there are three homophone pairs: *si*, meaning either ‘die’ or ‘four’, *sng*, meaning ‘play’ and ‘compute’, and *kai*, meaning ‘correct’ and ‘exceed’. They are all heterographic homophones, and thus are represented by different characters. The first meaning of each pair all belong to the citation group, and the second meaning all belong to the sandhi group. Figure 3 presents the predicted contours of these homophones obtained from the *word* model. It can be seen

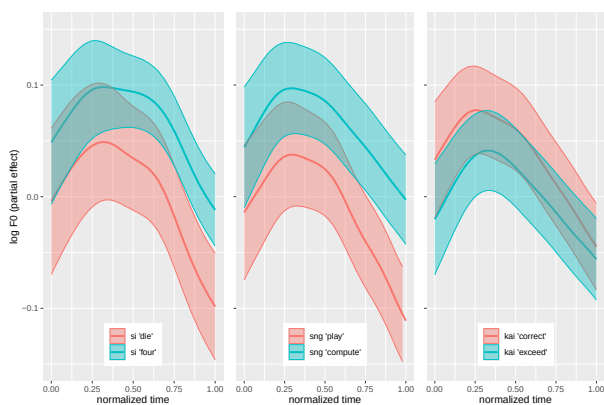


Figure 3: The predicted tonal contours of three heterographic homophone pairs. The red and green lines plot the contours of the citation and sandhi words respectively.

that homophones’ tonal realizations share more similarities than non-homophones, as presented in Figure 2. Nonetheless, even though homophones have exactly the same segmental makeup, their tonal contours still exhibit subtle distinction. For *si*, the meaning of the sandhi word (‘four’) seems to have a higher pitch register than that of the citation word (‘die’). The same holds for *sng*, as the contour of the sandhi word (‘compute’) is above that of the citation word (‘play’). The trend, however, is reversed for *kai*. Pitch register tends to be lower for the sandhi word than the citation word, although the difference is not as prominent. This pattern of results therefore illustrates that word meaning, among other differences encompassed in words, have a non-negligible effect on tonal realizations.

4. Discussion

This study investigated the variation in high-falling tone realizations in Taiwanese spontaneous speech. When comparing its sandhi and citation realizations, we found that the two are not identical. This finding is much in line with [3] and [4], who also reported incomplete neutralization for sandhi and citation realizations. In addition to that, our results further showed an interaction between neutralization and prosody. That is, how sandhi and citation tones differ is depending on prosodic positions. In general, the sandhi-citation difference is enlarged at prosodic boundaries than at pre-boundary positions. One possible explanation is that different boundary tone could have also influenced the realizations of sandhi and citation tones. Although it is not yet clear how intonation pattern affects tonal realizations in Taiwanese [18], the discrepancy found in the present study is likely to have resulted from other prosody-related factors.

The novel finding of this study is the effect of word meaning on tonal realizations. When the model has access to the word identity of the tokens, it is able to predict tonal contours with higher precision. This suggests that tonal realizations vary across words, and that every word has its own specific tonal contour. To make sure that the effect of word pertains to its meaning, we further examined the tonal realizations of homophones. It was found that homophone pairs, despite having the same segmental makeup, still differ in the fine-grained details of their tonal realizations. The current finding fits well with previous studies that demonstrate the intricate relationship between word meaning and phonetic realizations [9, 10, 20]. However, it

seems that the difference in tonal realization seems to be more subtle than other phonetic differences reported in previous homophone studies. We suspect that this is possibly due to the coarse meaning distinction in this study. As mentioned above, all homophones in this study are heterographic. However, characters are admittedly a poor representation of word meaning, as the majority of Taiwanese characters have more than one meanings (or senses). Take *kai* ‘correct’ (citation realization, Figure 3) for example. In addition to the meaning of correcting, according to the Dictionary of Frequently-Used Taiwanese [21], it has four other different senses, including ‘quitting’, ‘changing’, ‘repenting’, and ‘times [classifier]’. Similarly for the sandhi counterpart of *kai*, in addition to the verb meaning of ‘exceeding’, its other senses include ‘extremely’, ‘spreading’, ‘talking nonsense’, and ‘covering’. Given the considerable diversity of meaning associated with both words, we might have simply compared two tonal contours, each averaged across a wide variety of different meanings. It is therefore likely that the true meaning-specific tonal difference has been masked by the grand average contours. Nevertheless, despite the imprecision of our current meaning distinction, the current results still demonstrate the potential of word meaning to account for tonal variation. More fine-grained annotations that distinguish different senses of the same word are required for future studies.

If word meaning does contribute to tonal variation, what does it tell us about incomplete neutralization, given that the comparison of sandhi and citation realizations inevitably involve different words? In this study we found that although tone type (citation vs. sandhi) can explain some variance in the data when the model is agnostic for word identity, it however did not improve model fit any further when word information was already in the model, at least for pre-boundary tokens. This suggests that the sandhi-citation distinction is no longer necessary once meaning-induced tonal variation is taken into account. In other words, word is a substantially more powerful predictor than tone type, and the finding of incomplete neutralization in previous studies could have resulted from overlooking the effect of word meaning on tonal realizations.

If incomplete neutralization can be reduced to the word effect, this might have suggested that sandhi and citation tones are actually fully neutralized, as any difference between them can be attributed to by-word tonal variation. This is indeed one possibility. However, we are cautious about drawing this conclusion, because so far we do not have a good definition for ‘word meaning’. For example, is the meaning of *apple* the same in compounds like *apple tree*, *apple pie*, and *apple juice*? How about sentences like *I ate the apple* and *The apple was eaten*, where the connotations might differ slightly depending sentence structure? Should contextual meaning as such be considered as part of word meaning? We do not have an answer to these questions yet. It thus remains an empirical question how much tonal variation can be accounted for by word meaning if we take “meaning” really seriously. The strong version of the hypothesis could be: given that words’ positional change in an utterance also lead to changes in their contextual meaning, even tone sandhi, alternation of tonal categories, might be accounted for by meaning variation resulted from positional change. If this turns out to be the case, it then suggests that the system of tone sandhi might not be needed at all, as word meaning is sufficient to predict how tone should be realized in different utterance positions. And if there is no tone sandhi, neutralization will no longer be an issue either. Since answering these questions undoubtedly requires more data and in-depth analyses, we leave the exploration of these interesting issues to future studies.

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6. References

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